

learnings into how their merchandising is designed and displayed in order to acquire customers. This technique is advantageous in the market research and customer services fields.

Speech recognition: Just a few years back, we used to type keywords into the Google search bar to get some results. In the last few years, major IT companies like Google, Apple and Amazon have already shifted to the use of NLP for voice-based search, providing customers a seamless experience. Many industries have seen massive growth in the use of natural language processing tools to enable effective speech recognition processes and enhance productivity.

Components of NLP

NLP has two basic components:

- Natural language understanding (NLU)
- Natural language generation (NLG)

Natural language understanding

NLU is naturally harder than NLG tasks. So let's understand the challenges a machine faces while trying to understand language. There is a lot of ambiguity while learning or trying to interpret a language. For example, consider the following sentence:

He is looking for a match.

Here, what do you understand by 'match'? It could mean a life partner or a cricket/football match.

Lexical ambiguity can occur when a word has more than one meaning and the sentence that it is used in can be interpreted differently, depending on its correct context. Lexical ambiguity can be resolved to some extent by using parts-of-speech tagging techniques. Let us look at another example:

The chicken is ready to eat.

Is the chicken ready to eat its food or is the chicken dish ready for someone else to eat? You never know.

Syntactical ambiguity means when there is more than one meaning to a sequence of words. It is also termed a grammatical ambiguity. Here is

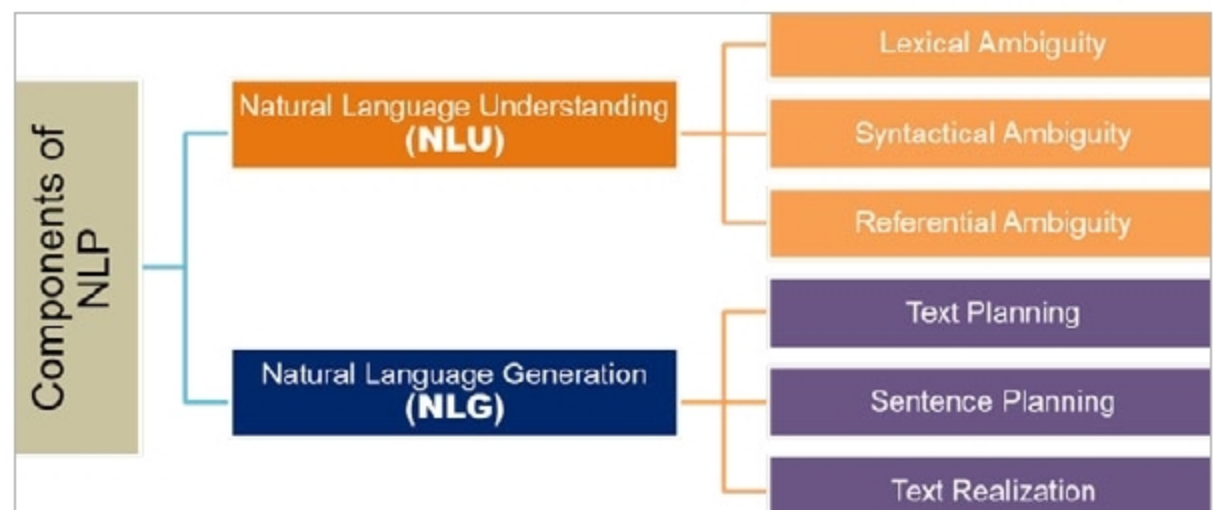


Figure 2: NLP components

another example:

Feluda met Topse and LalMohan babu. They went to a restaurant.

In this case, does 'they' refer to Topse and LalMohan babu or to all three of them?

Referential ambiguity happens when a text mentions an entity (something/ someone) and then refers to it again, possibly in a different sentence, using another word. This is the common case of pronouns causing ambiguity, when it is not clear which of the many nouns in a sentence is being referred to.

Natural language generation

This is the process of producing meaningful phrases and sentences in the form of natural language from some internal representation.

It involves:

- *Text planning*, which includes retrieving the relevant content from the knowledge base.
- *Sentence planning*, which includes choosing the required words, forming meaningful phrases and setting the tone of the sentence.
- *Text realisation*, which is mapping the sentence plan into the sentence structure.

Five common NLP terms

Tokenisation: The process of breaking down a paragraph of text into smaller chunks such as words or sentences is called tokenisation. A token is a single entity that is a building block for a sentence or a paragraph. This allows the

computer to deal with words in a more manageable form.

Using Python NLTK (Natural Language Tool Kit), we can easily perform tokenisation as shown below.

```
from nltk.tokenize import sent_tokenize
text="Hello friends!. Good Morning!
Today we will learn Natural Language
Processing. It is very interesting"
tokenized_text=sent_tokenize(text)
print(tokenized_text)
```

OUTPUT

```
['Hello friends!.', 'Good Morning!',
'Today we will learn Natural Language
Processing.', 'It is very interesting']
```

```
from nltk.tokenize import word_tokenize
tokenized_word=word_tokenize(text)
print(tokenized_word)
```

OUTPUT

```
['Hello', 'friends', '!', '.', 'Good',
'Morning', '!', 'Today', 'we', 'will',
'learn', 'Natural', 'Language',
'Processing', '.', 'It', 'is', 'very',
'interesting']
```

Stopwords removal: Stopwords are considered as noise in the text. A text may contain stopwords such as is, am, are, this, a, an, the, etc. We don't want these words to take up space in our database, or take up valuable processing time. They can be removed easily by storing a list of words that you may consider to be stopwords. NLTK has a list of stopwords stored in 16 different